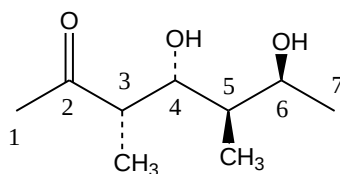


ACML 100 Practice problems

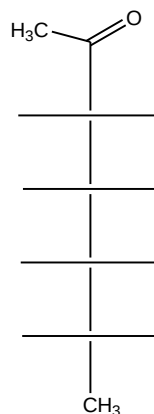
1. The concentration of X dissolved in chloroform is 6.15 g per 100 mL of solution. (a) A portion of this solution in a 5-cm polarimeter tube causes rotation of -1.2° . Calculate the specific rotation of X. (b) Predict the observed rotation if the same solution were placed in a 10-cm tube. (c) Predict the observed rotation if 10 mL of the solution were diluted to 20 mL and placed in a 5 cm tube.

2. The optical rotation of sugar is used in industry as a quick way to check the concentration of sugar solution. If 20 g of cane sugar dissolved in water to make up 100 mL of solution and placed in a tube of 40 cm long rotates the plane of polarized light $+53.2^\circ$ at 20°C . What is the concentration of another solution of sugar, measured at the same temperature and in the same polarimeter tube, if the optical rotation is $+13.3^\circ$.

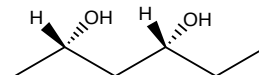
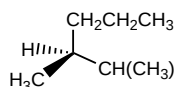
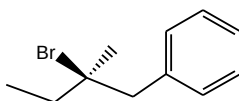
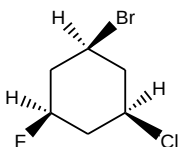
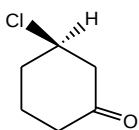
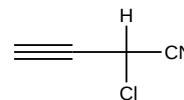
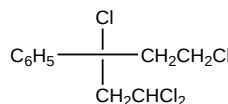
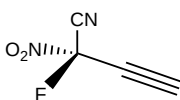
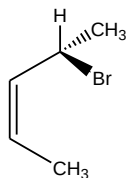
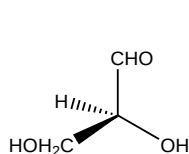
3. Assign absolute configuration at each of the chiral carbons in the given molecule (**B**). Convert the structure to Fisher projection in the template provided.



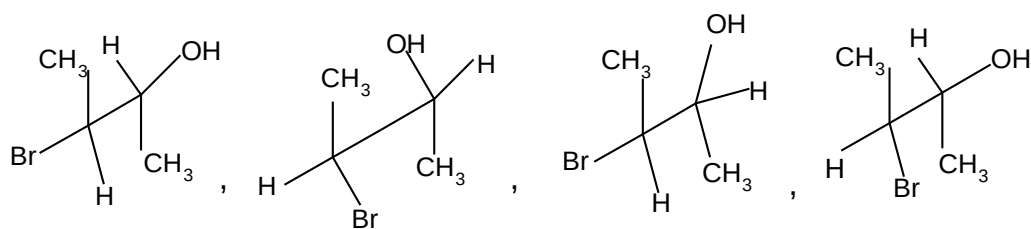
B



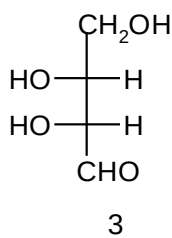
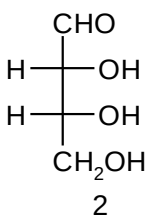
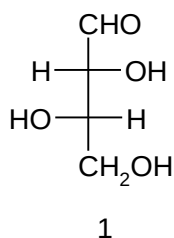
4. Assign R/S configuration to the stereocentres in the following molecules.



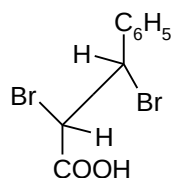
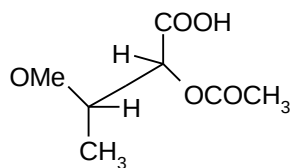
5. Which prefix *threo* or *erythro* is correctly applied to the following structures?



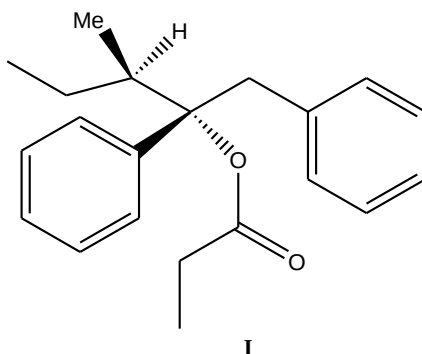
6. Write the Sawhorse and Newman structures for the following compounds and indicate whether any of these structures are superimposable.



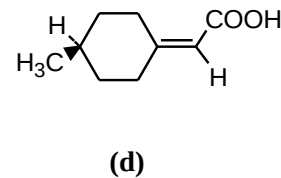
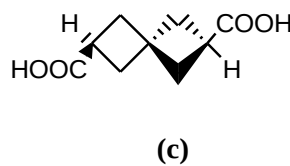
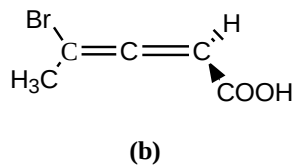
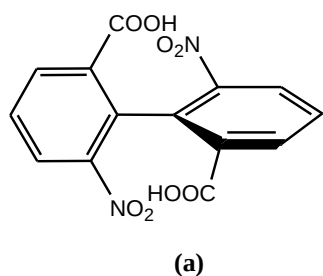
7. Convert the following structure to corresponding Fischer projection formulae.



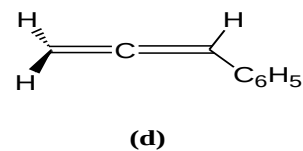
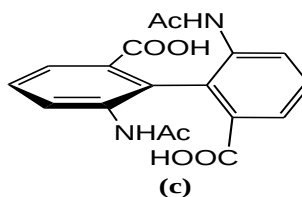
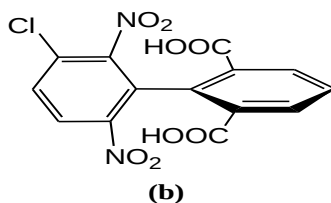
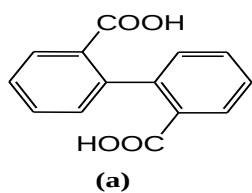
8. Darvon (**1**) is a painkiller. Assign the absolute configurations for the chiral centres if any. Draw the most stable sawhorse and Fischer projections of the molecule.



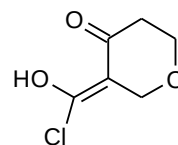
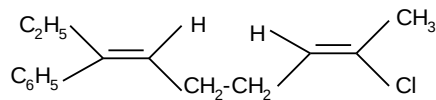
9. Assign absolute configuration to the following compounds.



10. Classify the following molecules as chiral or achiral.



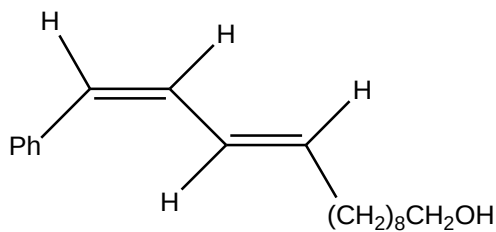
11. (a) Assign *E* or *Z* to the following compounds.



(b) Write all stereoisomers of octa-2,5-diene and assign configurational symbol to them.

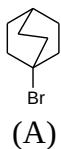
12. What is the configuration of the molecule given below?

(i) 10*Z*, 12*E* (ii) 10*E*, 12*Z* (iii) 10*E*, 12*E* (iv) 10*Z*, 12*Z*



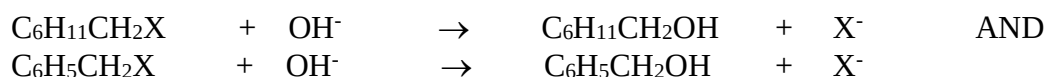
13. The relative rates of formation of ethers from sodium ethoxide and ethyl, propyl, isopropyl, neopentyl bromides are equal to 100, 28, 3 and 4×10^{-4} respectively. How will you account for these results?

14. The bicyclic bromide A resists hydrolysis by both S_N1 and S_N2 pathways. Why?



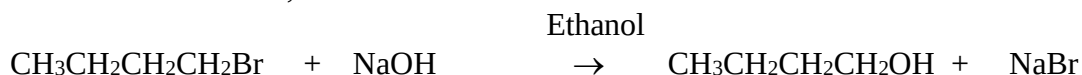
15. An aromatic dibromide $C_7H_6Br_2$ reacts with aqueous NaOH to give C_7H_7BrO which is insoluble in the reaction medium. Give the structure of both starting material and product and explain the insolubility of the product formed.

16. The reaction given below both react similarly at the X substituted carbon to give primary alcohols. However, they do so by different mechanism.



Explain why this is so and give the mechanism for each of the reaction. Describe also the experiment you will perform to differentiate the two mechanisms which are operative in this case (same experiment applied to both reactions).

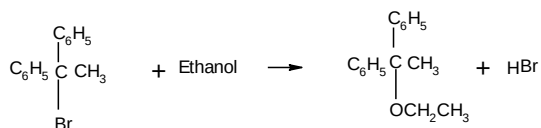
17. For the S_N2 reaction,



Answer the following questions:

- What is the rate expression for the reaction?
- Draw an energy diagram for the reaction. Label all parts. You may assume that the products are of lower energy than the starting reagents.
- What will be the effect on the rate of reaction if the concentration of *n*-butyl bromide is doubled?
- What will be the effect on the rate of reaction if the concentration of the NaOH is halved?
- How will the rate of the S_N2 reaction change if the solvent is changed to 80% ethanol and 20% water

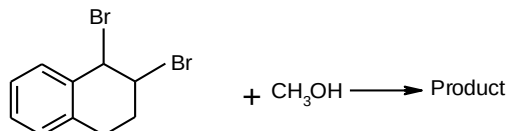
18. For S_N1 reaction,



- What is the rate expression for the reaction

- (b) Draw an energy diagram for the reaction. Label all parts. You may assume that the products are of lower in energy than the starting reagents.
- (c) What will be the effect on the rate of the reaction on doubling the initial concentration of 1-bromo-1,1-diphenylethane.
- (d) Will the rate of the $\text{S}_{\text{N}}1$ reaction change significantly if some water is added to the solvent ethanol.

19. When the following dibromo compound is dissolved in methanol, one of the bromine atoms is replaced by a methoxy group. Predict what the structure of the product of this reaction and why this one product is formed



20. Draw the structure of the bridged intermediate expected if each of the following were to act as a neighbouring group. To what class of compounds does each intermediate belong?

- (a) $-\text{N}(\text{CH}_3)_2$; (b) $-\text{SCH}_3$; (c) $-\text{OH}$; (d) $-\text{Br}$; (e) $-\text{C}_6\text{H}_5$; (f) $-\text{C}_6\text{H}_4\text{OCH}_3$ -*p*; (g) $-\text{CH}=\text{CHR}$

21. Give reasons in each of the following observations:

- (a) Sodium α -bromo propionate [$R\text{-CH}_3\text{CH}(\text{Br})\text{COONa}$] undergoes hydrolysis with retention of configuration.
- (b) NaOH readily converts *trans* 2-chloro cyclohexanol into cyclohexene oxide, but converts *cis* isomer into entirely different products.
- (c) $\text{CH}_3\text{CHOHCH}_2\text{SC}_2\text{H}_5$ or $\text{CH}_3\text{CH}(\text{SC}_2\text{H}_5)\text{CH}_2\text{OH}$ on reaction with HCl give the same product $\text{CH}_3\text{CHClCH}_2\text{SC}_2\text{H}_5$.
- (d) On treatment with aq. HBr both *cis* and *trans* 2-bromocyclopentanol are converted into the same product (Write the structure of the product also).

22. Account for the following observations:

- (a) Dehydrobromination of *meso* and *racemic* 1,2-dibromo-1,2-diphenylethane give *cis* and *trans* 1-bromo-1,2-diphenylethene respectively.
- (b) The relative rates of bimolecular dehydrohalogenation of *trans* 4-*ter*.butylcyclohexylbromide, *cis* 4-*ter*.butylcyclohexylbromide and cyclohexylbromide at 75°C are 0 , 7.1×10^{-3} and 2.4×10^{-3} respectively.

23. Each of the following alcohols can react with acid to form an alkene, (a) Predict the alkenes formed in these E1 reactions (Assume that no rearrangements occur). If several alkenes are possible, indicate which should predominate. (b) Which of the intermediate cations might be able to form a more stable cation, assuming that only one 1,2-shift is involved.

- (a) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- (b) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- (c) $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$
- (d) $\text{CH}_3\text{CH}(\text{CH}_3)\text{C}(\text{CH}_3)_2\text{OH}$
- (e)

